

Question ID d6456c7a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: d6456c7a

- A certain park has an area of **11,863,808** square yards. What is the area, in square miles, of this park? (**1 mile = 1,760 yards**)
- A. **1.96**
 - B. **3.83**
 - C. **3,444.39**
 - D. **6,740.8**

ID: d6456c7a Answer

Correct Answer:
B

Rationale

Choice B is correct. Since **1** mile is equal to **1,760** yards, **1** square mile is equal to **1,760²**, or **3,097,600**, square yards. It's given that the park has an area of **11,863,808** square yards. Therefore, the park has an area of **(11,863,808 square yards) $\left(\frac{1 \text{ square mile}}{3,097,600 \text{ square yards}}\right)$** , or $\frac{11,863,808}{3,097,600}$ square miles. Thus, the area, in square miles, of the park is **3.83**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the square root of the area of the park in square yards, not the area of the park in square miles.

Choice D is incorrect and may result from converting **11,863,808** yards to miles, rather than converting **11,863,808** square yards to square miles.

Question Difficulty:
Hard

Question ID 11b06e35

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 11b06e35

The density of a certain solid substance is **813** kilograms per cubic meter. A sample of this substance is in the shape of a cube, where each edge has a length of **0.60** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A. **176**
- B. **488**
- C. **1,355**
- D. **3,764**

ID: 11b06e35 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of **0.60** meters. Therefore, the volume of the sample is **0.60³**, or **0.216**, cubic meters. It's also given that the sample has a density of **813** kilograms per **1** cubic meter. Therefore, the mass of this sample is **(0.216 cubic meters) ($\frac{813 \text{ kilograms}}{1 \text{ cubic meter}}$)**, or **175.608** kilograms. Rounding this mass to the nearest whole number gives **176** kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is **176**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 20b69297

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 20b69297

Anita created a batch of green paint by mixing 2 ounces of blue paint with 3 ounces of yellow paint. She must mix a second batch using the same ratio of blue and yellow paint as the first batch. If she uses 5 ounces of blue paint for the second batch, how much yellow paint should Anita use?

- A. Exactly 5 ounces
- B. 3 ounces more than the amount of yellow paint used in the first batch
- C. 1.5 times the amount of yellow paint used in the first batch
- D. 1.5 times the amount of blue paint used in the second batch

ID: 20b69297 Answer

Correct Answer:

D

Rationale

Choice D is correct. It’s given that Anita used a ratio of 2 ounces of blue paint to 3 ounces of yellow paint for the first batch. For any batch of paint that uses the same ratio, the amount of yellow paint used will be $\frac{3}{2}$, or 1.5, times the amount of blue paint used in the batch. Therefore, the amount of yellow paint Anita will use in the second batch will be 1.5 times the amount of blue paint used in the second batch.

Alternate approach: It’s given that Anita used a ratio of 2 ounces of blue paint to 3 ounces of yellow paint for the first batch and that she will use 5 ounces of blue paint for the second batch. A proportion can be set up to solve for x, the amount of yellow paint she will use for the second batch: $\frac{2}{3} = \frac{5}{x}$. Multiplying both sides of this equation by 3 yields $2 = \frac{15}{x}$, and multiplying both sides of this equation by x yields $2x = 15$. Dividing both sides of this equation by 2 yields $x = 7.5$. Since Anita will use 7.5 ounces of yellow paint for the second batch, this is $\frac{7.5}{5} = 1.5$ times the amount of blue paint (5 ounces) used in the second batch.

Choices A, B, and C are incorrect and may result from incorrectly interpreting the ratio of blue paint to yellow paint used.

Question Difficulty:

Hard

Question ID 7d093333

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 7d093333

The area of a rectangular region is increasing at a rate of **250** square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use **1 meter = 3.28 feet**.)

- A. **0.39**
- B. **1.27**
- C. **13.67**
- D. **23.24**

ID: 7d093333 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that **1 meter = 3.28 feet**. It follows that **1² square meter = 3.28² square feet**, or **1 square meter = 10.7584 square feet**. Since **1 hour = 60 minutes**, it follows that **250** square feet per hour is equivalent to $\left(\frac{250 \text{ square feet}}{1 \text{ hour}}\right)\left(\frac{1 \text{ square meter}}{10.7584 \text{ square feet}}\right)\left(\frac{1 \text{ hour}}{60 \text{ minutes}}\right)$, or $\frac{250 \text{ square meters}}{645.504 \text{ minutes}}$, which is approximately **0.3873** square meters per minute. Of the given choices, **0.39** is closest to **0.3873**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 21e539a0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 21e539a0

A landscaper uses a hose that puts $88x$ ounces of water in a bucket in $5y$ minutes. Which expression represents the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate?

- A. $\frac{9x}{440}$
- B. $\frac{440x}{9}$
- C. $\frac{5x}{792}$
- D. $\frac{792x}{5}$

ID: 21e539a0 Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that a hose puts $88x$ ounces of water in a bucket in $5y$ minutes. Therefore, the rate at which the hose puts water in the bucket, in ounces per minute, can be represented by the expression $\frac{88x}{5y}$. Let w represent the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate. It follows that the rate at which the hose puts water in the bucket, in ounces per minute, can be represented by the expression $\frac{w}{9y}$. The expressions $\frac{88x}{5y}$ and $\frac{w}{9y}$ represent the same rate, so it follows that $\frac{88x}{5y} = \frac{w}{9y}$. Multiplying both sides of this equation by $9y$ yields $\frac{792xy}{5y} = w$, or $\frac{792x}{5} = w$. Therefore, the number of ounces of water the hose puts in the bucket in $9y$ minutes can be represented by the expression $\frac{792x}{5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID c2e7fa6d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c2e7fa6d

For an electric field passing through a flat surface perpendicular to it, the electric flux of the electric field through the surface is the product of the electric field’s strength and the area of the surface. A certain flat surface consists of two adjacent squares, where the side length, in meters, of the larger square is **3** times the side length, in meters, of the smaller square. An electric field with strength **29.00** volts per meter passes uniformly through this surface, which is perpendicular to the electric field. If the total electric flux of the electric field through this surface is **4,640 volts · meters**, what is the electric flux, in **volts · meters**, of the electric field through the larger square?

ID: c2e7fa6d Answer

Correct Answer:
4176

Rationale

The correct answer is **4,176**. It’s given that the side length of the larger square is **3** times the side length of the smaller square. This means that the area of the larger square is **3²**, or **9**, times the area of the smaller square. If the area of the smaller square is represented by ***x***, then the area of the larger square can be represented by **9*x***. Therefore, the flat surface of the two adjacent squares has a total area of ***x* + 9*x***, or **10*x***. It’s given that an electric field with strength **29.00** volts per meter passes uniformly through this surface and the total electric flux of the electric field through this surface is **4,640 volts · meters**. Since it’s given that the electric flux is the product of the electric field’s strength and the area of the surface, the equation **29.00(10*x*) = 4,640**, or **290*x* = 4,640**, can be used to represent this situation. Dividing each side of this equation by **290** yields ***x* = 16**. Substituting **16** for ***x*** in the expression for the area of the larger square, **9*x***, yields **9(16)**, or **144**, square meters. Since the area of the larger square is **144** square meters, the electric flux, in **volts · meters**, of the electric field through the larger square can be determined by multiplying the area of the larger square by the strength of the electric field. Thus, the electric flux is **(144 square meters)($\frac{29.00 \text{ volts}}{\text{meter}}$)**, or **4,176 volts · meters**.

Question Difficulty:
Hard

Question ID 50b99b2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 50b99b2d

Objects R and S each travel at a constant speed. The speed of object R is half the speed of object S. Object R travels a distance of $4x$ inches in y seconds. Which expression represents the time, in seconds, it takes object S to travel a distance of $24x$ inches?

- A. $12y$
- B. $3y$
- C. $16y$
- D. $6y$

ID: 50b99b2d Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that object R travels a distance of $4x$ inches in y seconds. This speed can be written as $\frac{4x \text{ inches}}{y \text{ seconds}}$. It's given that the speed of object R is half the speed of object S. It follows that the speed of object S is twice the speed of object R, which is $2\left(\frac{4x \text{ inches}}{y \text{ seconds}}\right)$, or $\frac{8x \text{ inches}}{y \text{ seconds}}$. Let n represent the time, in seconds, it takes object S to travel a distance of $24x$ inches. The value of n can be found by solving the equation $\frac{8x \text{ inches}}{y \text{ seconds}} = \frac{24x \text{ inches}}{n \text{ seconds}}$, which can be written as $\frac{8x}{y} = \frac{24x}{n}$. Multiplying each side of this equation by ny yields $8xn = 24xy$. Dividing each side of this equation by $8x$ yields $n = 3y$. Therefore, the expression $3y$ represents the time, in seconds, it takes object S to travel a distance of $24x$ inches.

Choice A is incorrect. This expression represents the time, in seconds, it would take object S to travel a distance of $24x$ inches if the speed of object R were twice, not half, the speed of object S.

Choice C is incorrect. This expression represents the time, in seconds, it takes object S to travel a distance of $128x$ inches, not $24x$ inches.

Choice D is incorrect. This expression represents the time, in seconds, it takes object R, not object S, to travel a distance of $24x$ inches.

Question Difficulty:
Hard

Question ID 5154615f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 5154615f

To study fluctuations in composition, samples of pumice were taken from **29** locations and cut in the shape of a cube. The length of the edge of one of these cubes is **3.000** centimeters. This cube has a density of **0.230** grams per cubic centimeter. What is the mass of this cube, in grams?

ID: 5154615f Answer

Correct Answer:
6.21

Rationale

The correct answer is **6.21**. It's given that the samples of pumice were cut in the shape of a cube. It's also given that the length of the edge of one of these cubes is **3.000** centimeters. Therefore, the volume of this cube is **(3.000 centimeters)³**, or **27** cubic centimeters. Since the density of this cube is **0.230** grams per cubic centimeter, it follows that the mass of this cube is $\left(\frac{0.230 \text{ grams}}{1 \text{ cubic centimeter}}\right)(27 \text{ cubic centimeters})$, or **6.21** grams.

Question Difficulty:
Hard

Question ID 61f61789

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 61f61789

To study the moisture content in a group of trees, samples from the trunk of each tree were taken from **25** trees and cut in the shape of a cube. The length of the edge of one of these cubes is **2.00** centimeters. If this cube has a mass of **2.56** grams, what is the density of this cube, in grams per cubic centimeter?

ID: 61f61789 Answer

Correct Answer:
0.32, 8/25

Rationale

The correct answer is **.32**. The volume of a cube is given by the formula $V = s^3$, where s is the length of an edge of the cube. It's given that each edge of the cube has a length of **2.00** centimeters. Substituting **2.00** centimeters for s in the formula $V = s^3$ yields $V = (2.00 \text{ centimeters})^3$, or $V = 8.00$ cubic centimeters. It's given that the cube has a mass of **2.56** grams. Dividing the mass, in grams, of the cube by the volume, in cubic centimeters, of the cube gives its density, in grams per cubic centimeters. Therefore, the density of the cube is $\frac{2.56 \text{ grams}}{8.00 \text{ cubic centimeters}}$, or **.32** grams per cubic centimeter. Note that .32 and 8/25 are examples of ways to enter a correct answer.

Question Difficulty:
Hard

Question ID c7c6445f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c7c6445f

- A certain town has an area of **4.36** square miles. What is the area, in square yards, of this town? (**1 mile = 1,760 yards**)
- A. **404**
 - B. **7,674**
 - C. **710,459**
 - D. **13,505,536**

ID: c7c6445f Answer

Correct Answer:
D

Rationale

Choice D is correct. Since the number of yards in **1** mile is **1,760**, the number of square yards in **1** square mile is $(1,760)(1,760) = 3,097,600$. Therefore, if the area of the town is **4.36** square miles, it is $4.36(3,097,600) = 13,505,536$, in square yards.

Choice A is incorrect and may result from dividing the number of yards in a mile by the square mileage of the town.

Choice B is incorrect and may result from multiplying the number of yards in a mile by the square mileage of the town.

Choice C is incorrect and may result from dividing the number of square yards in a square mile by the square mileage of the town.

Question Difficulty:
Hard

Question ID 7d721177

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 7d721177

The density of a certain type of wood is **353** kilograms per cubic meter. A sample of this type of wood is in the shape of a cube and has a mass of **345** kilograms. To the nearest hundredth of a meter, what is the length of one edge of this sample?

- A. 0.98
- B. 0.99
- C. 1.01
- D. 1.02

ID: 7d721177 Answer

Correct Answer:
B

Rationale

Choice B is correct. It’s given that the density of a certain type of wood is **353** kilograms per cubic meter (kg/m^3), and a sample of this type of wood has a mass of **345 kg**. Let x represent the volume, in m^3 , of the sample. It follows that the relationship between the density, mass, and volume of this sample can be written as $\frac{353 \text{ kg}}{1 \text{ m}^3} = \frac{345 \text{ kg}}{x \text{ m}^3}$, or $353 = \frac{345}{x}$. Multiplying both sides of this equation by x yields $353x = 345$. Dividing both sides of this equation by **353** yields $x = \frac{345}{353}$. Therefore, the volume of this sample is $\frac{345}{353} \text{ m}^3$. Since it’s given that the sample of this type of wood is a cube, it follows that the length of one edge of this sample can be found using the volume formula for a cube, $V = s^3$, where V represents the volume, in m^3 , and s represents the length, in m, of one edge of the cube. Substituting $\frac{345}{353}$ for V in this formula yields $\frac{345}{353} = s^3$. Taking the cube root of both sides of this equation yields $\sqrt[3]{\frac{345}{353}} = s$, or $s \approx 0.99$. Therefore, the length of one edge of this sample to the nearest hundredth of a meter is **0.99**.

Choices A, C, and D are incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 8637294f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 8637294f

If $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$, what is the value of n ?

ID: 8637294f Answer

Correct Answer:
.0625, 1/16

Rationale

The correct answer is **.0625**. It's given that $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$. The equation $\frac{4a}{b} = 6.7$ can be rewritten as $(4)(\frac{a}{b}) = 6.7$. Dividing both sides of this equation by **4** yields $\frac{a}{b} = 1.675$. The equation $\frac{a}{bn} = 26.8$ can be rewritten as $(\frac{a}{b})(\frac{1}{n}) = 26.8$. Substituting **1.675** for $\frac{a}{b}$ in this equation yields $(1.675)(\frac{1}{n}) = 26.8$, or $\frac{1.675}{n} = 26.8$. Multiplying both sides of this equation by n yields **1.675 = 26.8n**. Dividing both sides of this equation by **26.8** yields $n = 0.0625$. Therefore, the value of n is **0.0625**. Note that .0625, 0.062, 0.063, and 1/16 are examples of ways to enter a correct answer.

Question Difficulty:
Hard

Question ID 3638f413

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3638f413

Jeremy deposited x dollars in his investment account on January 1, 2001. The amount of money in the account doubled each year until Jeremy had 480 dollars in his investment account on January 1, 2005. What is the value of x ?

ID: 3638f413 Answer

Rationale

The correct answer is 30. The situation can be represented by the equation $x(2^4) = 480$, where the 2 represents the fact that the amount of money in the account doubled each year and the 4 represents the fact that there are 4 years between January 1, 2001, and January 1, 2005. Simplifying $x(2^4) = 480$ gives $16x = 480$. Therefore, $x = 30$.

Question Difficulty:
Hard

Question ID e7133228

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: e7133228

The speed of a vehicle is increasing at a rate of **7.3** meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth? (Use **1 mile = 1,609 meters**.)

- A. 0.3
- B. 16.3
- C. 195.8
- D. 220.4

ID: e7133228 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the speed of a vehicle is increasing at a rate of **7.3** meters per second squared. It's given to use **1 mile = 1,609 meters**. There are **60** seconds in **1** minute; therefore, **60²** or **3,600** seconds squared is equal to **1** minute squared. It follows that the rate of **7.3** meters per second squared is equivalent to $\left(\frac{7.3 \text{ meters}}{1 \text{ second squared}}\right)\left(\frac{1 \text{ mile}}{1,609 \text{ meters}}\right)\left(\frac{3,600 \text{ seconds squared}}{1 \text{ minute squared}}\right)$, or approximately **16.33 miles per minute squared**. The rate, in **miles per minute squared**, rounded to the nearest tenth is **16.3**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 69f6717f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 69f6717f

A sample of oak has a density of **807** kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of **0.90** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A. **588**
- B. **726**
- C. **897**
- D. **1,107**

ID: 69f6717f Answer

Correct Answer:
A

Rationale

Choice A is correct. It’s given that the sample is in the shape of a cube with edge lengths of **0.9** meters. Therefore, the volume of the sample is **0.90³**, or **0.729**, cubic meters. It’s also given that the sample has a density of **807** kilograms per **1** cubic meter. Therefore, the mass of this sample is **0.729 cubic meters** $\left(\frac{807 \text{ kilograms}}{1 \text{ cubic meter}}\right)$, or **588.303** kilograms. Rounding this mass to the nearest whole number gives **588** kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is **588**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard